AS YOU SIT DOWN

Get your workspace open

THREE THINGS

- VS Code open on your **ECON209** folder
- In `labs/project/`, three folders: `data`, `code`, `output`
- A new notebook, **`lab5.ipynb`**, in `labs/project/code/`

We'll download the data together.

IF SOMETHING ISN'T WORKING

- Sit next to someone whose setup runs
- Follow along now, fix yours after

APPLIED ECONOMETRICS · RICE UNIVERSITY

Lab Five

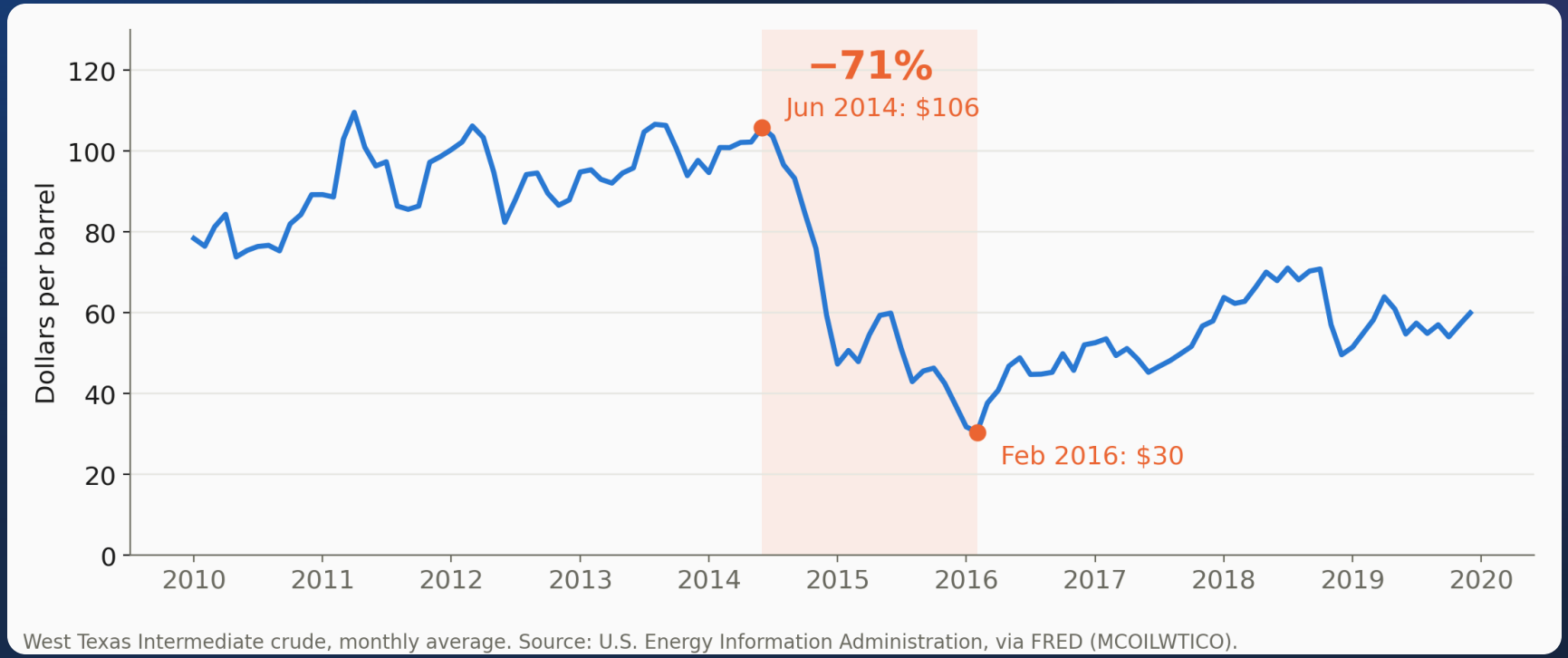
Getting data from a website, making dummies, summary statistics, and reading least squares output.

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Teaching Assistant

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Questions, any time

Sewall 207
Mondays

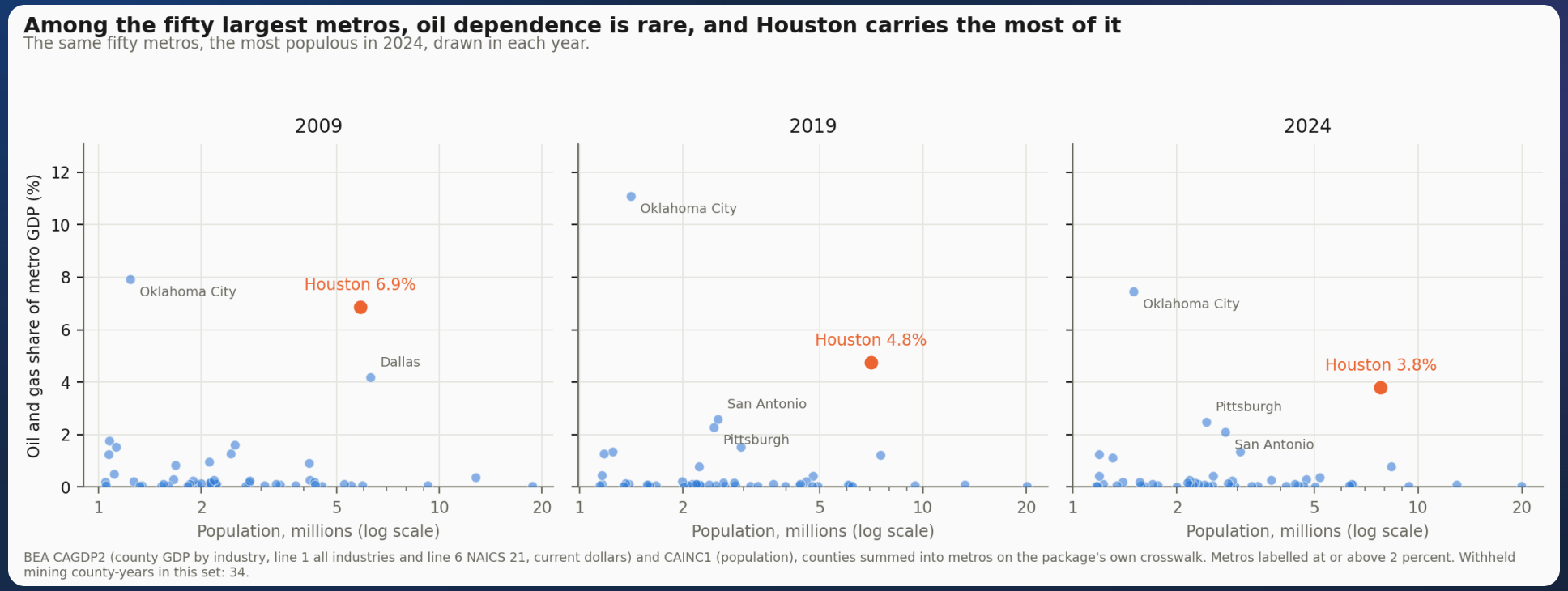
Oil prices fell 71% in 2014–16



West Texas Intermediate crude, monthly average. Source: U.S. Energy Information Administration, via FRED (MCOILWTICO).

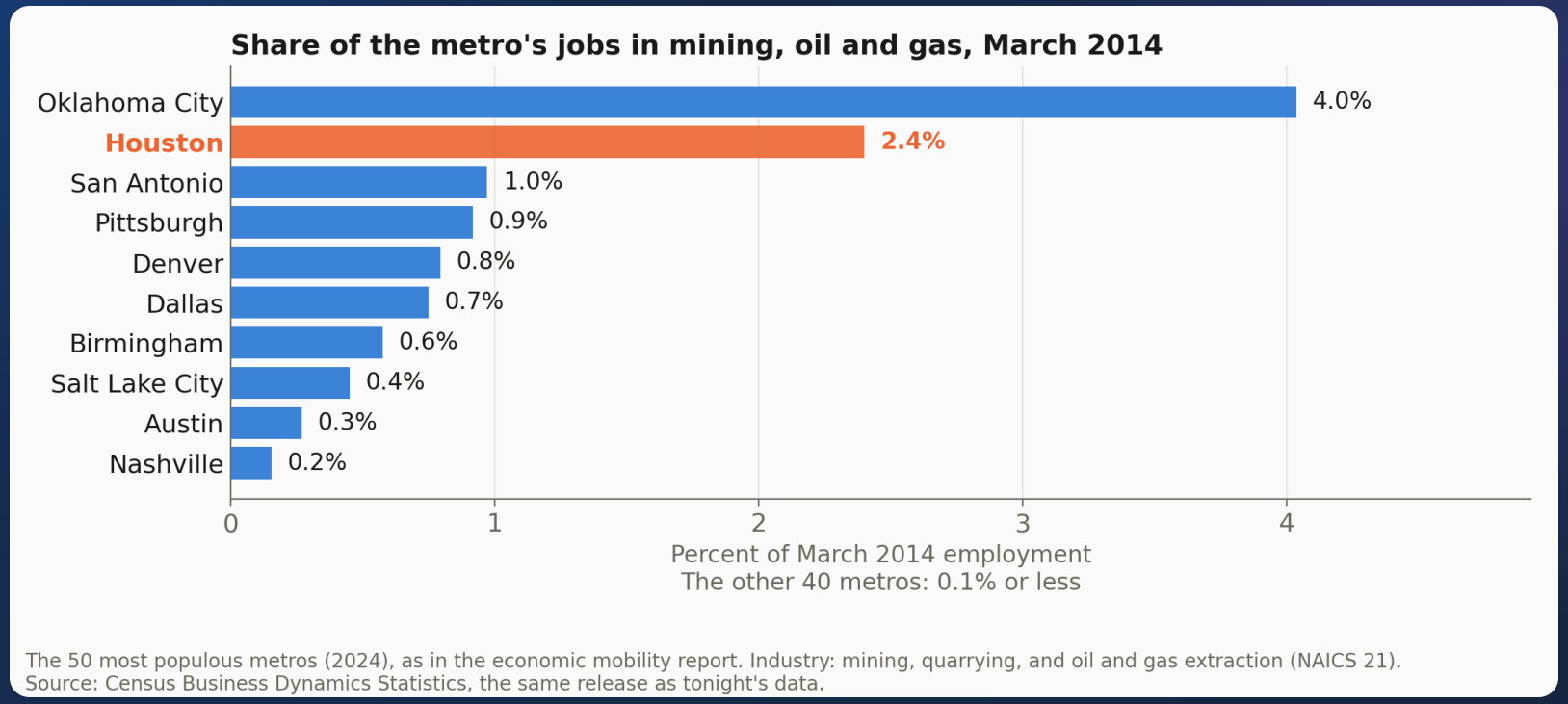
The economic mobility report asks what that shock did to Houston's jobs, its incomes, and who could reach them.

Oil's share of GDP in the 50 largest metros



Houston: 6.9% of its output in 2009, 4.8% in 2019, 3.8% in 2024. Most large metros are near zero.

Houston ranks second in oil jobs



The 50 most populous metros (2024), as in the economic mobility report. Industry: mining, quarrying, and oil and gas extraction (NAICS 21).
Source: Census Business Dynamics Statistics, the same release as tonight's data.

The industry is mining and oil and gas extraction. Houston's refining and petrochemicals count in other industries.

TONIGHT

Four parts, and how we'll work

1. Get the data from a website
2. Make it usable: load, keep 2013, label
3. A dummy and summary statistics
4. Least squares, and reading the output

HOW WE'LL WORK

- I'll ask Claude for each step's code, live in VS Code. You can too
- Each slide: what, why, and what to be careful about
- Each ends with a **check**: numbers your code should reproduce
- Complete work may not be correct work

PART ONE · THE DATA

Business Dynamics Statistics, by metro

- From the Census Bureau, and one of the economic mobility report's sources
- Public, free, no account needed: the kind of data your project will use
- One row per metro area per year, 1978–2023

TONIGHT'S QUESTION

- **y:** `job_creation_rate`, 2013: jobs added by new and growing workplaces (% of jobs)
- **x:** `very_large`, 1 for metros with 1 million or more jobs
- In 2013, did the biggest metros create jobs at a higher rate?

To study Houston, the report compares it with other large metros. Tonight we look at all of them in 2013, before the shock, to see how metro job markets worked.

Getting the file

1. On census.gov, search Business Dynamics Statistics
2. Click the result Business Dynamics Statistics (BDS) Data
3. Click the Datasets tab
4. Under One-Way Datasets, click MSA [6MB]
5. Move `bds2023_msa.csv` into `labs/project/data/`

BE CAREFUL

- If it opens as text: go back, right-click the link, **Save Link As...** (Safari: **Download Linked File**)
- Don't open and re-save it in Excel

PART ONE · READ THE DOCUMENTATION

What the columns mean, and where it came from

TECHNICAL DOCUMENTATION → CODEBOOK AND GLOSSARY

- `emp`: paid employees on the payroll in the pay period including March 12
- `msa`: a code for each metropolitan *and* micropolitan area
- `S`: a number suppressed for data quality

WRITE IT DOWN · YOUR NOTEBOOK'S FIRST CELL

- The source: Census Bureau, Business Dynamics Statistics
- The file, and how you found it
- The date you downloaded it

So you, or anyone, can get the same file again.

PART TWO

Load it and take a first look

WHAT WE'RE DOING

- Read the CSV file into a table
- Tell pandas that S means missing
- Look at the first rows, the years, and the number of areas

WHY

- Check that you have what you think you have before you use it

BE CAREFUL

- A column that holds an S is read as text, not numbers, unless you say S means missing
- The path starts from the notebook's folder: `../data/`
- Don't open and re-save the file in Excel

Check: `42,550` rows and `26` columns · years `1978–2023` · `925` areas

PART TWO

Keep one year: 2013

WHAT WE'RE DOING

- Keep only the rows for 2013
- That leaves one row per metro

WHY

- A snapshot before the shock: how metro job markets worked in 2013

BE CAREFUL

- Afterwards there should be exactly one row per metro
- The year is a number: compare with 2013, not "2013"

Check: 925 rows, one for each area

PART TWO

Add the metro names

WHAT WE'RE DOING

- Search census.gov for **delineation files** and open **Delineation Files**
- Under **Core based statistical areas (CBSAs)...**, click **Jul. 2023**
- Join it to the metros by code

WHY

- Codes are hard to read and check; the list also says which areas are metropolitan

BE CAREFUL

- Take the first Jul. 2023 link, not Principal cities
- Excel file: title rows above the table, notes below
- One row per county: keep one row per area
- After joining, count metros with no name, and check one you know

Check: `0` metros without a name · `26420` is Houston

PART TWO

The analysis sample

WHAT WE'RE DOING

- Rules for 2013: employment known; metropolitan areas only; 100,000+ jobs

WHY

- A written rule lets anyone rebuild the sample
- The names file's labels make the metro rule possible

BE CAREFUL

- Apply the rules one at a time; count after each
- A different cutoff gives a different sample: write it down

Check: 925 → 387 → 171 metros

PART THREE

Making a dummy variable

WHAT WE'RE DOING

- `very_large`: 1 for metros with 1 million or more jobs in 2013, 0 for the rest

WHY

- A dummy turns a group into a number least squares can use

BE CAREFUL

- Count the 1s
- The cutoff is a choice: write it down

Check: 22 metros with `very_large` = 1

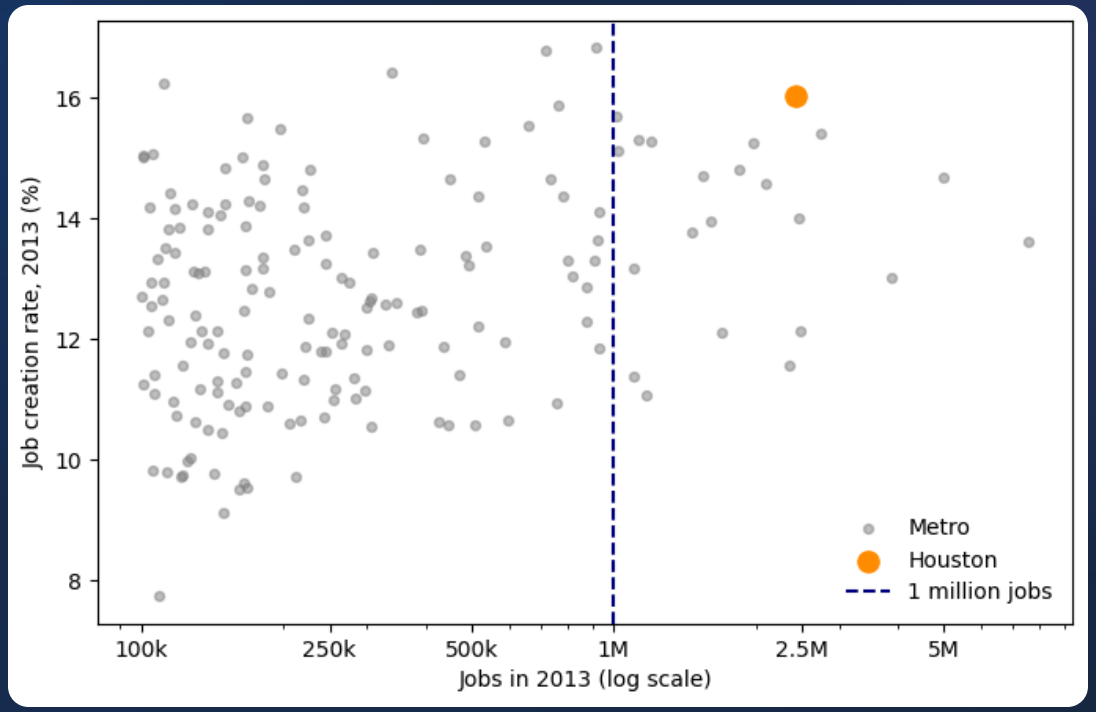
PART THREE

Summary statistics

	count	mean	std	min	max
job_creation_rate	171.0	12.723	1.781	7.735	16.825
very_large	171.0	0.129	0.336	0.000	1.000

- **count:** every variable has all 171 metros
- **The mean of a dummy is a share:** 12.9% of these metros have 1 million or more jobs
- **min and max:** job creation rates run from 7.7% to 16.8%. Look at the range before you fit

Where does Houston fall?



Each dot is a metro in 2013. Right of the dashed line: 1 million or more jobs. Houston (orange) created jobs at 16.0%, the highest of the biggest metros.

PART FOUR · WHY REGRESS

Did big metros create jobs faster?

- Houston is one of the 22 metros with 1 million or more jobs
- If size predicts job creation, comparisons with Houston must account for it

$$y = \beta_0 + \beta_1 \cdot \texttt{very_large} + u$$

Least squares answers it: the best straight line for predicting y from x.

$$y_i = \beta_0 + \beta_1 x_i + u_i$$

FOR EACH METRO $i = 1, \ldots, 171$

- **y_i:** job creation rate, 2013 (%)
- **x_i:** $\texttt{very_large}_i$, 1 if metro i has 1 million+ jobs
- **β_0, β_1:** the intercept and slope to estimate
- **u_i:** everything else about metro i

LEAST SQUARES

- Picks β_0, β_1 to make the total of squared misses as small as possible
- $\beta_0 + \beta_1 x$ is the best linear predictor of y given x, $E^*[y \mid x]$
- Robust (HC1) standard errors: how precisely we know them

$$job_creation_rate_i = \beta_0 + \beta_1 \cdot very_large_i + u_i$$

Dep. Variable:	job_creation_rate	R-squared:	0.069
Model:	OLS	Adj. R-squared:	0.063
Method:	Least Squares	F-statistic:	16.15
Date:	Mon, 05 Oct 2026	Prob (F-statistic):	8.79e-05
Time:	17:16:07	Log-Likelihood:	-334.75
No. Observations:	171	AIC:	673.5
Df Residuals:	169	BIC:	679.8
Df Model:	1		
Covariance Type:	HC1		

CHECK FIRST

The y you meant, all 171 metros, and HC1

R-SQUARED

The share of the variation in y the line explains

SKIP FOR NOW

The F-statistic, AIC, BIC, and the block at the bottom

PART FOUR · THE COEFFICIENTS

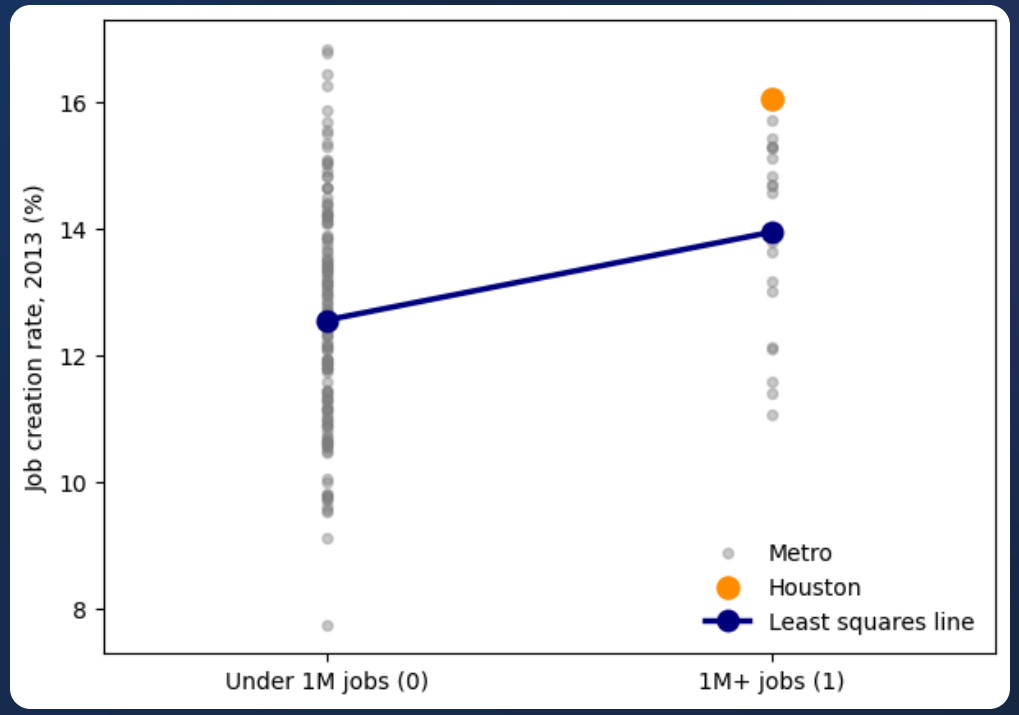
predicted job_creation_rate = 12.5443 + 1.3909 · very_large

	coef	std err	z	P>|z|	[0.025	0.975]
Intercept	12.5443	0.144	87.129	0.000	12.262	12.827
very_large	1.3909	0.346	4.019	0.000	0.713	2.069

- **Intercept, 12.5443:** average rate for metros under 1 million jobs
- **very_large, 1.3909:** the biggest metros' rate was 1.39 points higher
- **95% interval, 0.713 to 2.069:** excludes 0, so we can rule out no difference

Check: group means 12.5443 (under 1M jobs) and 13.9352 (1M+), difference 1.3909

The line runs through the two group means



$\beta_0 = 12.5443$ is the mean on the left. $\beta_1 = 1.3909$ is how much the line rises to the mean on the right.